Anisotropic spacing of drumlins: transverse quasi-regularity at the bedform scale versus along-flow clustering in the BRITICE record

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Abstract
Drumlins are widely interpreted as the emergent product of a subglacial bed-flow instability that selects a characteristic wavelength, and large-sample studies report that drumlin fields are statistically "patterned" rather than random. Because drumlins are strongly elongate, however, the sense in which they are regularly spaced has not been resolved. We analyse 41 702 drumlin outlines from the BRITICE v2 compilation of Great Britain, reduce each to a centroid with a flow-parallel long-axis azimuth and a width, segment the data into coherent single-flow fields, and decompose the spacing into directions parallel and transverse to ice flow. Isotropic nearest-neighbour statistics are *clustered*, not regular (Clark–Evans $R \leq 1$ in all nine fields analysed, median 0.89): at the centroid level drumlins form along-flow trains. The regularity is purely *transverse*: in flow-aligned coordinates the across-flow pair correlation shows hard-core exclusion and a characteristic spacing $\lambda_{\perp} \approx 250\,\mathrm{m}$, of the same order as the drumlin width ($\sim 250\,\mathrm{m}$). Compared with a width-matched hard-rod (Tonks-gas) null, the transverse first-neighbour spacing is largely accounted for by steric exclusion, with only weak longer-range transverse periodicity beyond it. The structure factor is dominated by field patchiness at low wavenumber and is featureless at the spacing scale, so the pattern is not hyperuniform. Drumlin "spacing regularity" is therefore an anisotropic, transverse, bedform-width-scale phenomenon rather than an isotropic lattice-like or hyperuniform order — a distinction that constrains how spacing statistics should be used to test bed-instability theories.

1 Introduction

Drumlins — streamlined hills of sediment formed beneath ice sheets and aligned with ice flow — are the most abundant subglacial bedform, yet their formation remains debated. A leading family of hypotheses holds that drumlins are an emergent instability of the coupled flow of ice and deforming till: a flat bed is unstable, and the fastest-growing mode selects a characteristic wavelength, so that bedforms should be quasi-regularly spaced (Hindmarsh, 1998; Fowler, 2000; Fowler & Chapwanya, 2014). Because the instability wavelength is predicted to set the bedform size as well as its spacing, regular spacing and a preferred size are expected to appear together.

Empirically, Clark et al. (2018b) examined 42 000 drumlins from Britain, Ireland, Canada and Norway with inhomogeneous spatial statistics and concluded that "patterning is a near-ubiquitous property of drumlins," with a regularity signal across length scales of roughly 100–1200 m. That study treated the spacing as an isotropic property. Drumlins are, however, strongly elongate (typical length-to-width ratios of 2–4), so a single "spacing" conflates two very different geometries: the along-flow arrangement of bedforms in trains and the across-flow arrangement of neighbouring trains. The instability wavelength of the theory is intrinsically *transverse* (incipient ridges grow across flow and are later dissected into individual drumlins), so a physically meaningful test of regular spacing must separate the two directions. It is also unknown whether any transverse spacing is independent of the bedform width, or simply reflects the fact that drumlins, being finite-width objects, cannot overlap side by side.

Here we use the BRITICE v2 drumlin compilation (Clark et al., 2018a) to decompose drumlin spacing into along-flow and transverse components, to test whether any transverse regularity exceeds simple steric (non-overlap) packing at the drumlin width, and to ask whether the pattern is hyperuniform — i.e. whether long-wavelength density fluctuations are suppressed (Torquato, 2018). The analysis uses only open data and standard, edge-corrected point-pattern and structure-factor statistics.

2 Data

The BRITICE v2 GIS database (Clark et al., 2018a) compiles glacial landforms of the last British–Irish Ice Sheet from more than 1800 sources, in the British National Grid (metres). We use the filtered *Subglacial lineations (polygons)* layer, whose attribute table classifies 41 702 features as drumlins (the more elongate mega-scale glacial lineations are mapped as crest-lines in a separate layer and are not used here). Each drumlin polygon is reduced to its area-weighted centroid; a principal-axis decomposition of the outline gives the long-axis azimuth (taken as the local ice-flow direction) and the width (minor-axis extent). Across the dataset the median width is $\sim$250 m and the median length $\sim$750 m, consistent with published drumlin dimensions.

3 Methods

Field segmentation. Drumlins occur in discrete fields separated by drumlin-free ground, and flow direction varies between fields; pooling them would superpose distinct processes. We link drumlins whose centroids lie within three median nearest-neighbour distances and take the connected components as fields. Inference is restricted to coherent fields with at least 300 drumlins and a high azimuth concentration (axial mean resultant length $\bar{R}_\theta \geq 0.72$), giving nine fields of 504–1547 drumlins.

Isotropic and directional pair statistics. For each field we compute the Clark–Evans nearest-neighbour index R (Clark & Evans, 1954) against a Monte-Carlo complete-spatial-randomness (CSR) null in the matched convex hull ($R > 1$ dispersed, $R < 1$ clustered). We then rotate the field into flow-aligned coordinates (long-axis azimuth along the y-axis) and compute directional pair correlations: the along-flow $g_\parallel$ and the transverse $g_\perp$, each accumulated from pairs lying within a thin slab (0.5 km) of the perpendicular coordinate so that the one-dimensional spacing in each direction is isolated. The isotropic $g(r)$ is also computed for reference.

Steric (hard-rod) null. To test whether transverse regularity exceeds simple non-overlap packing, we compare $g_\perp$ to an equilibrium one-dimensional hard-rod (Tonks-gas) null in which each slab is repopulated with the same number of rods of diameter equal to the measured drumlin width, placed at random subject to non-overlap. The envelope of this null is the spacing expected from steric exclusion alone at the observed density and width.

Structure factor. The radially averaged structure factor $S(k) = N^{-1}\left|\sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}\right|^2$, evaluated at the wavevectors of the bounding window, tests for hyperuniformity: a vanishing $S(k \to 0)$ would indicate suppressed long-wavelength density fluctuations (Torquato, 2018), whereas a packing peak with finite $S(k \to 0)$ indicates a liquid-like spacing.

4 Results

4.1 Isotropic statistics: clustering, not regularity

At the centroid level the drumlin fields are *clustered.* The Clark–Evans index is ≤ 1 in all nine fields (median $R = 0.89$; Fig. 3b), and the isotropic $g(r)$ rises to 2–3 at small separations and decays monotonically (Fig. 2, centre) — the signature of clustering, with no hard-core and no packing peak. The along-flow pair correlation $g_{\parallel}$ is also > 1 at short range (Fig. 2, centre): drumlins are arranged in along-flow trains. Thus the naive expectation of isotropic regular spacing is not met; isotropic nearest-neighbour regularity is absent.

4.2 Transverse quasi-regularity at the bedform scale

The regularity is entirely transverse. In flow-aligned coordinates the across-flow pair correlation $g_{\perp}$ shows clear hard-core exclusion at small separation and a characteristic-spacing peak (Fig. 2, right). The transverse spacing is $\lambda_{\perp} \approx 250\,\mathrm{m}$ (median over the nine fields), of the same order as the median drumlin width ($\sim 250\,\mathrm{m}$); across fields $\lambda_{\perp}$ tracks the drumlin width to within a factor of about two (Fig. 3a). The pattern is therefore strongly anisotropic: clustered along flow, quasi-regular across flow, at a transverse scale set by the bedform width.

4.3 Is the transverse spacing more than steric packing?

Because $\lambda_{\perp}$ is close to the drumlin width, much of the transverse "regularity" could simply reflect the impossibility of side-by-side overlap. Comparison with the width-matched hard-rod null bears this out for the first neighbour: the observed $g_{\perp}$ peak falls at or near the steric envelope (Fig. 2, right), so the nearest-neighbour transverse spacing is largely accounted for by steric exclusion at the bedform width. Beyond the first neighbour the observed $g_{\perp}$ retains weak oscillations that persist slightly further than the steric null, hinting at a modest degree of genuine transverse ordering, but the effect is small and we do not interpret it as strong periodicity. In statistical-mechanical terms the across-flow arrangement is, to first order, a one-dimensional *Tonks gas* — a line of impenetrable rods of the bedform width — dominated by steric exclusion rather than by a system-wide dynamical wavelength, with the weak residual ordering beyond the first neighbour the only signal not reproduced by that null.

4.4 The pattern is not hyperuniform

The radially averaged structure factor is dominated at low wavenumber by the large-scale patchiness of each field (a strong rise as $k \to 0$) and is featureless — close to the random value of unity — at the wavenumbers corresponding to the drumlin spacing. There is no suppression of long-wavelength density fluctuations; the drumlin point pattern is not hyperuniform.

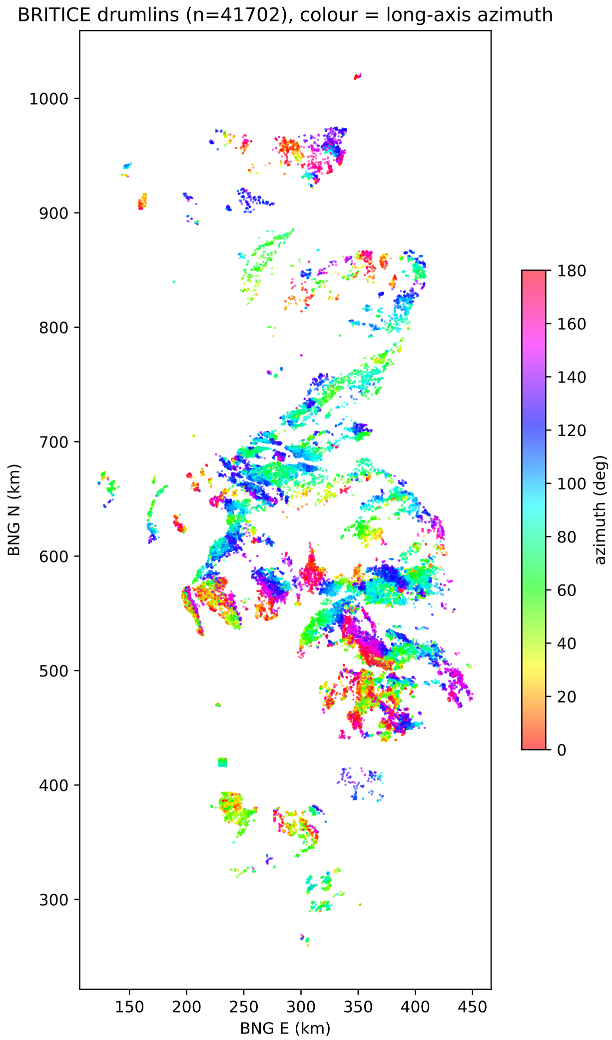

Figure 1: BRITICE v2 drumlins of Great Britain (41 702 centroids), coloured by long-axis (ice-flow) azimuth. Drumlins occur in discrete fields, each with a locally coherent flow direction; field boundaries coincide with abrupt azimuth changes.

5 Discussion

An anisotropic, bedform-scale spacing. The single "regular spacing" attributed to drumlins resolves, on direction-resolved analysis, into two opposite behaviours: along-flow *clustering* (bedforms in trains) and across-flow *quasi-regularity* at a scale comparable to the drumlin width. The previously reported isotropic regularity (Clark et al., 2018b) is consistent with this picture — an isotropic statistic mixes a clustered and a regular direction and returns an intermediate, weakly patterned signal — but the direction-resolved view localises the regularity to the transverse direction and ties its scale to the bedform width.

Implications for instability theories. That the transverse spacing scales with the drumlin width is exactly what an instability theory predicts, since a single wavelength sets both the size and the cross-flow spacing of the bedforms (Hindmarsh, 1998; Fowler, 2000; Fowler & Chapwanya, 2014;

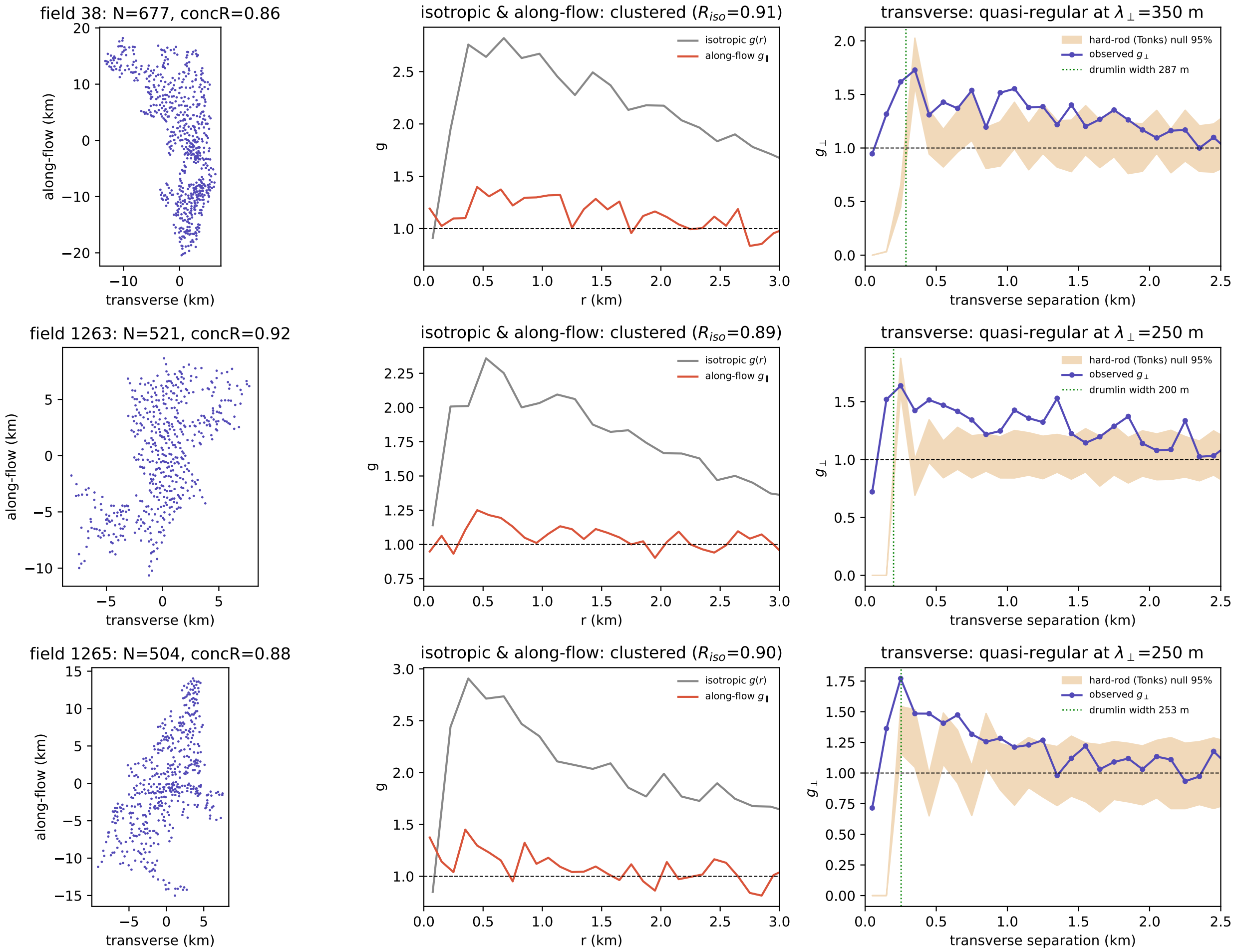

Figure 2: Three representative coherent drumlin fields. Left: centroids in flow-aligned coordinates. Centre: the isotropic $g(r)$ (grey) and along-flow $g_{\|}$ (red) are both > 1 at short range (clustering; $R_{\text{iso}} < 1$). Right: the transverse $g_\perp$ (purple) shows hard-core exclusion and a characteristic spacing $\lambda_\perp$ near the drumlin width (green line); the shaded band is the 95% envelope of a width-matched hard-rod (steric) null.

Clark et al., 2009). The same coincidence, however, means that transverse spacing close to the width cannot by itself discriminate a pattern-forming instability from simple steric packing of finite-width objects: at the first neighbour the two are observationally degenerate, as the hard-rod comparison shows. Evidence for a genuine wavelength must come from the *longer-range* transverse order, which we find to be only weak. Spacing regularity alone is thus a weak test of the instability hypothesis; size–spacing covariation and the persistence of transverse order over several bedform widths are more diagnostic. This steric dominance of the mature pattern does not in itself invalidate bed-instability models. A fluid-mechanical instability may still govern the nucleation of bedforms and the early selection of a wavelength; but as a field matures, non-linear saturation and the geometric crowding of finite-width bedforms can overprint that initial pattern, so that the preserved record — such as the BRITICE compilation of relict drumlins — is effectively truncated at the scale of the bedforms' own width. The instability hypothesis is therefore better tested on active or immature fields, or through size–spacing covariation, than on mature spacing statistics.

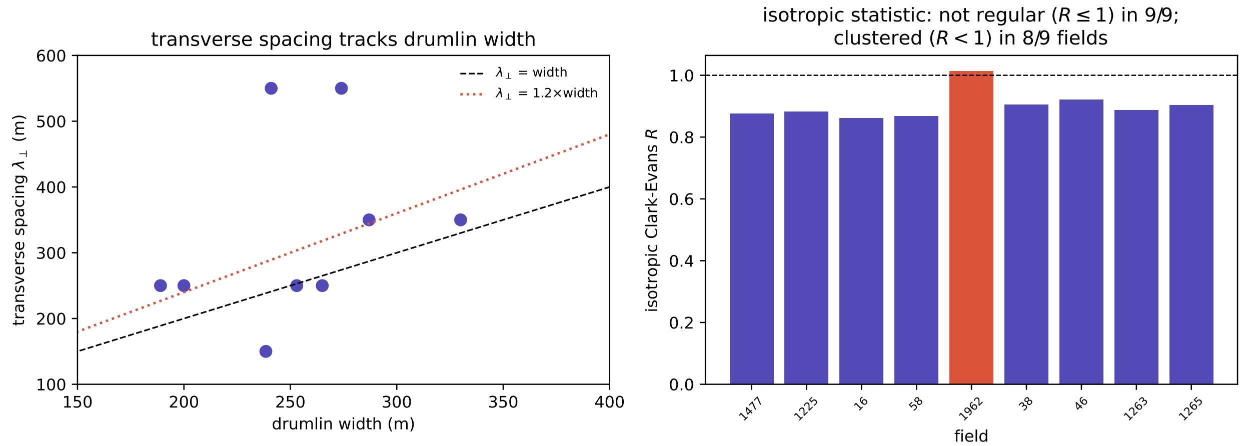

Figure 3: (a) Transverse spacing $\lambda_\perp$ versus drumlin width across the nine fields; $\lambda_\perp$ is of the same order as the width (dashed: equality; dotted: 1.2×). (b) Isotropic Clark–Evans R per field: $R \leq 1$ in all nine fields (clustered, $R < 1$, in eight), confirming that isotropic regularity is absent.

Why anisotropic: shear flow versus buoyancy. The anisotropy itself is informative. Drumlins form within a continuous, directional basal shear flow, which breaks the symmetry of the bed: sediment is advected and reworked along flow, favouring along-flow continuity of bedforms (trains), much as in subaqueous and aeolian dunes, while the cross-flow direction is free to take up a width-scale spacing. This differs from spacing systems driven by a vertical buoyancy (Rayleigh–Taylor) instability with no imposed horizontal direction — for example salt structures, which can develop an isotropic, hard-core, liquid-like regular spacing in map view (without reaching long-range, lattice-like order; Chen, 2026). Whether an external flow direction is imposed may thus be a first-order control on whether a geological spacing pattern is isotropically regular or anisotropically clustered.

Not a hyperuniform pattern. Unlike some self-organised geological patterns, drumlin centroid fields show no suppression of long-wavelength density fluctuations: the structure factor is patchiness-dominated at low wavenumber and random at the spacing scale. The order in drumlin fields is local and transverse, not long-range.

6 Conclusions

- Resolved by direction, BRITICE drumlin spacing is strongly anisotropic: *clustered* along ice flow (bedforms in trains; isotropic Clark–Evans $R \leq 1$ in all nine fields) and *quasi-regular* across flow.
- The transverse spacing $\lambda_\perp \approx 250$ m is of the same order as the drumlin width; at the first neighbour it is largely accounted for by steric (non-overlap) packing, with only weak longer-range transverse order beyond it.
- The pattern is not hyperuniform; long-wavelength density fluctuations are not suppressed.
- Isotropic spacing regularity is therefore a weak diagnostic of bed-instability theories for drumlins; transverse, direction-resolved statistics and size–spacing covariation are required.

Data availability

Drumlin outlines are from the BRITICE Glacial Map v2 (Clark et al., 2018a), freely available under a Creative Commons Attribution licence from the University of Sheffield. All analysis code, the derived drumlin-centroid table and the figure-generation scripts are openly available at `https://github.com/Ruqing1963/drumlin-anisotropic-spacing`. Analysis used only open scientific Python (NumPy, SciPy, Matplotlib).

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